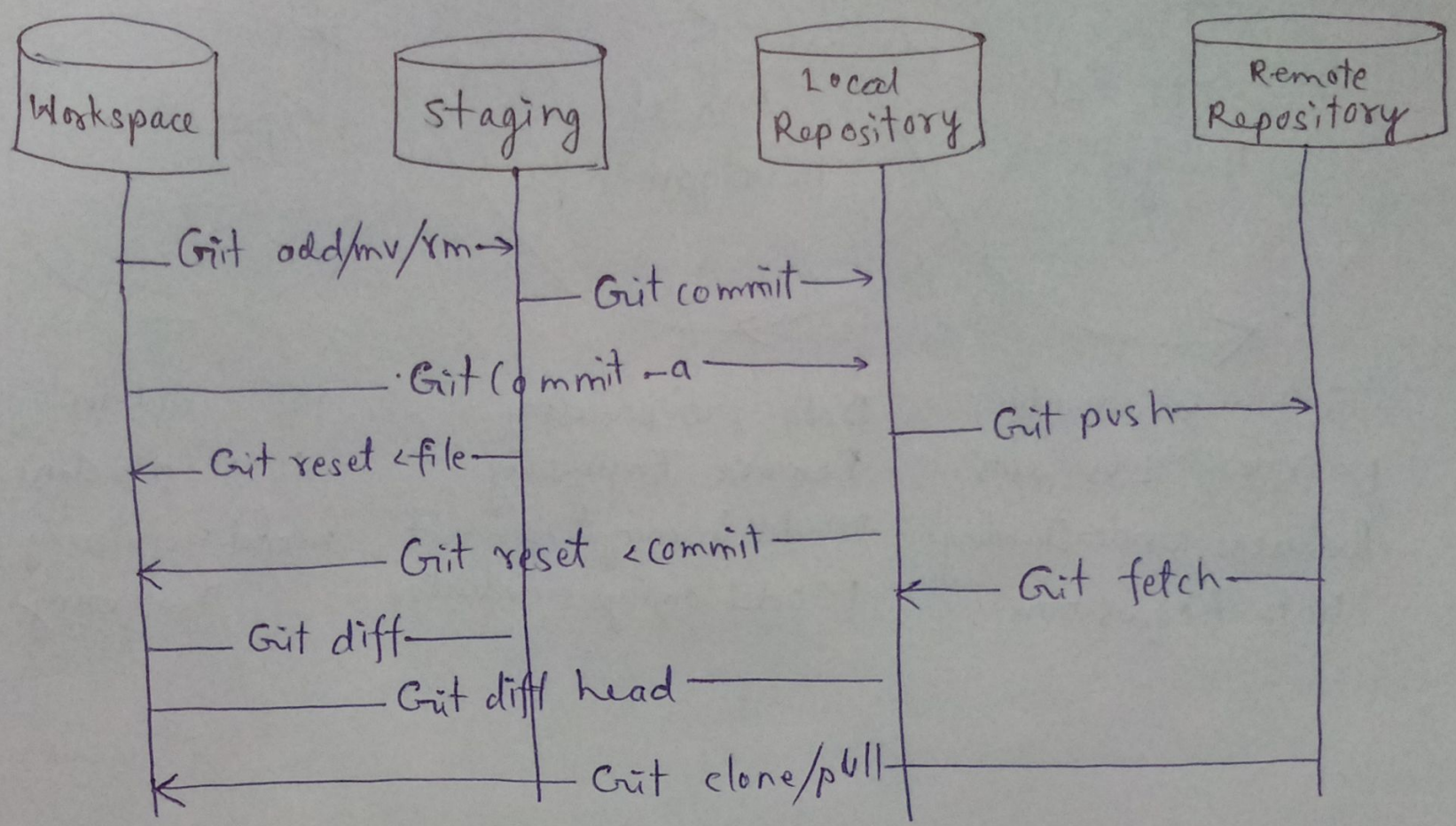
Date- 30 Apr 2026

Git and GitHub



Basic required before moving to MLOps

- Git is a VCS/SCM
- What is the need? Why should we learn?
  - Bug fixing and version control
  - Collaboration
  - Non-linear development
- Download/install from git-scm.com

Basic Setup

- Defining the user name
  git config --global user.name Vikash
- Defining the mail
  git config --global user.email vikashdas770@gmail.com

- check your user name
  git config --global user.name
- check your email
  git config --global user.email

Basic Commands
- git init (to initialize a git repo)
- git status (tells you current status of git repo)
- git add filename.py or $\boxed{\text{git add .}}$ → for all files
  (move work to staging area)
- git commit -m "commit message"
- Create a .gitignore file and add files/folder that doesn't need tracking
- git restore --staged <file> (to the role back)

Seeing Commits
- git log (see details for all commits)
- git log --online (1 liner details for all commits)
- git log --stat (details of changes on commit level)
- git log -p (see code changes on commit level)
- git show <6 digit sha id> (see changes on specific commit)
- git diff (changes made within staging area before commit)
- git checkout <6 digit sha id> (roll back to original)

git checkout master (Resume state)

Branching and Merging
- git branch <name> <sha id> (create a new branch)
- git branch (list all branches)
- git checkout <branch> (switch to another branch)
- git log --online --all (see commit of all branches)
- git log --online -all --graph
  (graph of commits from all branches)
- git branch -d/-D <branch> (deletes a branch)
- git merge <branch> (run from master/main)

Working with remote Repo

- git remote add origin <url> (declaring remote repo)
- git push origin master/main (sends local repo to origin)
- git pull origin master (pull code from remote repo)
- git clone <url> (download/copy remote repo)

Understanding software Versions → V.1.0.0

- git tag -a v1.0.0 <sha (optional)>
  Vim editor
  Fn+ insert - write a message
  :wq - save and quit
- git log (check if tagging was done successfully)
- git tag -d v1.0.0 (delete the mentioned tag)